HW 3

1. (a) $X_1 \sim \text{Pois}(\lambda)$

$$E[X_1] = \sum_{k=0}^{\infty} k * \frac{\lambda^k}{k!} * e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} * e^{-\lambda}$$

$$= \lambda \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} * e^{-\lambda} = \lambda$$

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} X_i \equiv \overline{X}_n.$$

(b) bias : $E_\lambda(\hat{\lambda}) - \lambda = E_\lambda\left(\frac{1}{n}\sum X_i\right) - \lambda = 0$

variance : $V(\hat{\lambda}) = V\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n} V(X_i) = \frac{1}{n}\lambda$

To derive the variance of X_1,

$$E[X_1(X_1-1)] = \sum_{k=0}^{\infty} k(k-1) * \frac{\lambda^k}{k!} * e^{-\lambda} = \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} * e^{-\lambda}$$

$$= \lambda^2 \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} * e^{-\lambda} = \lambda^2$$

$$E[X_1^2] = E[X_1(X_1-1) + X_1] = \lambda^2 + \lambda$$

$$V(X_1) = E[X_1^2] - (E[X_1])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$MSE(\hat{\lambda}) = V(\hat{\lambda}) + (\text{bias}(\hat{\lambda}))^2$$

$$= \frac{1}{n}\lambda$$

(c) $\eta = p(X_1 > 5) = 1 - p(X_1 \le 4)$

$$= 1 - \sum_{k=0}^{4} \frac{\lambda^k}{k!} e^{-\lambda}$$

by plug-in principle,

$$\hat{\eta} = 1 - \sum_{k=0}^{4} \frac{\hat{\lambda}^k}{k!} e^{-\hat{\lambda}}$$

where $\hat{\lambda}$ is the MoM estimator $\bar{X}_n$.

(d) $\mathcal{L}_n(\lambda) = \prod_{i=1}^{n}\left(\frac{\lambda^{X_i}}{X_i!} \cdot e^{-\lambda}\right) \propto e^{-\lambda n}\,\lambda^{\sum X_i}$

$\ell_n(\lambda) = \text{const} - \lambda n + (\sum X_i)\log\lambda$

F.O.C. $\dot{\ell}_n(\lambda) = -n + (\sum X_i)\frac{1}{\lambda} = 0$

$\hat{\lambda} = \bar{X}_n$

S.O.C. $\ddot{\ell}_n(\lambda) = -(\sum X_i)\lambda^{-2} < 0$

then. $\hat{\lambda}_{MLE} = \bar{X}_n$

(e) by equi-variance / invariance of MLE

$\widehat{\log(\lambda)}_{MLE} = \log(\hat{\lambda}_{MLE}) = \log(\bar{X}_n)$

(f). by LLN, $\bar{X}_n \xrightarrow{p} \mathbb{E}[X_1] = \lambda_0$

note $\log(\cdot)$ is continuous.

by continuous mapping thm, $\log(\bar{X}_n) \xrightarrow{p} \log(\lambda_0)$.

Q2.

(a). $\mathbb{E}[X_1] = \int_0^\infty x\,\lambda e^{-\lambda x}\,dx = -\int_0^\infty x\,de^{-\lambda x}$

$= \int_0^\infty e^{-\lambda x}\,dx = \frac{1}{\lambda}$

set $\frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{\hat{\lambda}}$ $\hat{\lambda}_{MOM} = (\bar{X}_n)^{-1}$

$$E_0(1/\hat{\lambda}_{mom}) = E_0(\bar{X}_n) = E_0 X_1 = \frac{a_0}{b_0}$$

$E_0[\cdot]$ denotes the expectation w.r.t. true data generating process.

$$E_0(X_1) = \int_0^\infty x \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx} dx$$

$$= \int_0^\infty \frac{b^a}{\Gamma(a)} x^{(a+1)-1} e^{-bx} dx$$

$$= \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+1)}{b^{a+1}} = \frac{a}{b}$$

(b). $$L_n(\lambda) = \prod_{i=1}^n (\lambda e^{-\lambda X_i}) = \lambda^n e^{-\lambda \sum X_i}$$

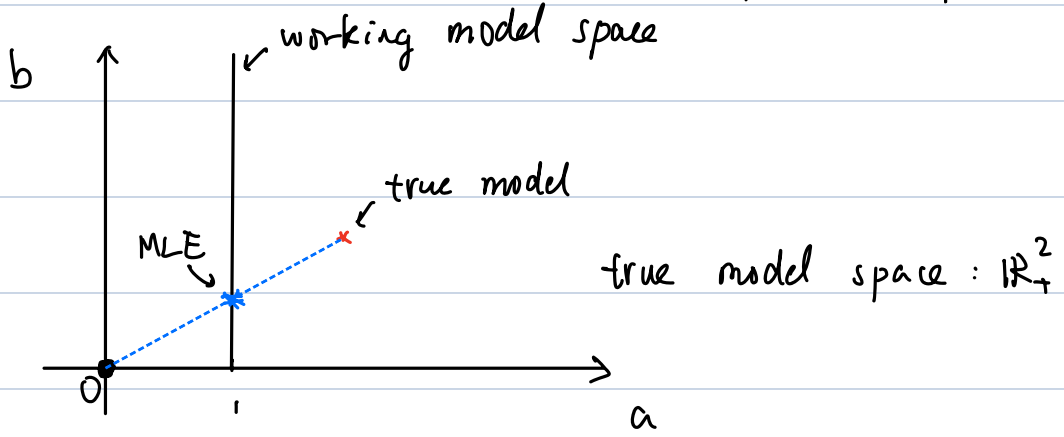
$$\ell_n(\lambda) = n \log(\lambda) - \lambda \sum X_i$$

F.O.C. $$\dot{\ell}_n(\lambda) = \frac{n}{\lambda} - \sum X_i = 0$$

S.O.C. $$\ddot{\ell}_n(\lambda) = -\frac{n}{\lambda^2} < 0$$

so $$\hat{\lambda}_{MLE} = (\bar{X}_n)^{-1}$$

(c). $Exp(\lambda) = Ga(1, \lambda)$. shape-rate parameterization.



only when true model $\in$ working model space, MLE is consistent.

MLE is minimizing $KL(Ga(a,b) \| Ga(1,\lambda))$ w.r.t. λ.

$$KL(Ga(a,b) \| Ga(1,\lambda)) = \int_0^\infty \log \frac{f_{a,b}(x)}{f_{1,\lambda}(x)} \, f_{a,b}(x) \, dx$$

$$= \text{const} - \mathbb{E}[\log f_{1,\lambda}(X_1)], \quad f_{1,\lambda}(x) = \lambda e^{-\lambda x}$$

$$= \text{const} + \mathbb{E}_0[\lambda X_1] - \log \lambda$$

$$= \text{const} + \frac{a}{b}\lambda - \log \lambda$$

$\tilde{\lambda} = \frac{b}{a}$ minimizes $KL(Ga(a,b) \| Ga(1,\lambda))$